

PARTICULAR SPECIFICATION P5A

APPENDIX P5A – TWRP – Hazardous Area Classification

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P5A.1 INTRODUCTION

This document has been prepared for PUB (the Board) by Jacobs Singapore Pte Ltd.

The purpose of this report is to assess and identify those areas within the Phase 1 Works at Tuas Water Reclamation Plant (TWRP) that are likely to have hazardous gases from Groups IIA and IIB and hazardous dusts as defined in IEC 60079-10 and to give structures, buildings and sub-structures a Hazardous Area designation in accordance with IEC 60079-10. This is to ensure that all plant installed in these areas are suitably rated to eliminate the potential risk of explosion.

NOTE: This classification is based on the design contained in the tender documents. Review will be required to assess any adjustments needed due to the particular equipment selected by the Contractor.

P5A.2 EXPLOSIVE ATMOSPHERES

P5A.2.1 Requirement for Area Classification

Many gases, vapours, mists and dusts encountered in industry are flammable and some of these occur in the water and wastewater industry. When ignited, they may burn readily and with considerable explosive force if mixed with air in appropriate proportions. It is often necessary to use mechanical and electrical apparatus in locations where such flammable materials may be present, and appropriate precautions must therefore be taken to ensure that all such apparatus is adequately protected in order to reduce the likelihood of ignition of any explosive atmosphere. When using electrical apparatus, potential ignition sources include electrical arcs and sparks, hot surfaces and, in certain circumstances with mechanical equipment sparks caused by friction.

In general, protection against mechanical and electrical ignition sources is achieved by implementing one of two practices:

- (a) Whenever practicable, the apparatus should be located outside hazardous areas; and
- (b) Alternatively, the apparatus must be designed, installed and maintained in accordance with measures recommended for the area in which the apparatus is to be located.

It is, therefore, necessary to identify areas within which such precautions are required. This is known as Hazardous Area Classification (HAC) or the zoning of hazardous areas. The purpose of area classification is to provide a basis for the correct selection, installation and location of electrical and non-electrical equipment in those areas. Areas must be classified depending on the properties of the flammable vapours, liquids, gases, mists, dusts when a combustible concentration is present.

The intent of the area classification is to reduce to an acceptable minimum level the probability of a flammable atmosphere and an ignition source occurring at the same time.

Equipment to be installed in hazardous zones shall be CE marked as complying with the ATEX European Directive 94/9/EU or IECEX, and shall comply with IEC 60079: Electrical apparatus for explosive gas atmospheres.

The result of the classification will be the division of the hazardous areas into zones.

P5A.3 METHODOLOGY FOR AREA CLASSIFICATION

This hazardous area classification uses the International Electrotechnical Commission (IEC) methodology and standards as defined by:

- 60079-10-1: *Explosive atmospheres - Part 10-1: Classification of areas - Explosive gas atmospheres standard*; and
- 60079-10-2: *Explosive atmospheres - Part 10-2: Classification of areas - Explosive dust atmospheres*.

These standards reference the Institute of Petroleum model code of safe practice in the petroleum industry; Part 15 - Area Classification Code for Installations Handling Flammable Fluids (IP15) as the supporting document. The terms used within IP15 are consistent with the usage and principles in IEC 60079-10.

Classification of the areas has been carried out in accordance with the requirements of IEC 60079-10. An analysis in accordance with this standard for the site is included.

IEC 60079 -10 states that “the area classification should be carried out by those who have knowledge of the properties of flammable materials, the process and the equipment, in consultation, as appropriate, with safety, electrical and other engineering personnel.”

P5A.3.1 Definitions

Non-hazardous (NH)	An area in which an explosive atmosphere is not expected to be present in quantities such as to require special precautions for the construction, installation and use of apparatus.
Hazardous area	An area in which an explosive atmosphere is present or may be expected to be present, in quantities such as to require special precautions for the construction, installation and use of apparatus.

Hazardous areas are classified into zones based on the frequency of occurrence and duration of an explosive gas atmosphere, as follows¹:

Zone 0	An area in which an explosive atmosphere consisting of a mixture with air of flammable substances in the form of gas, vapour or mist is present continuously or for long periods or frequently.
Zone 1	An area in which an explosive atmosphere consisting of a mixture with air of flammable substances in the form of gas, vapour or mist is likely to occur in normal operation occasionally.
Zone 2	An area in which an explosive atmosphere consisting of a mixture with air of flammable substances in the form of gas, vapour or mist is not likely to occur in normal operation but, if it does occur, will persist for a short period only.
Zone 20	An area in which an explosive dust atmosphere, in the form of a cloud of dust in air, is present continuously, or for long periods or frequently.

¹ As per IEC60079-10-1.1 Electrical apparatus for explosive gas atmospheres. Classification of hazardous areas.

Zone 21 An area in which an explosive dust atmosphere, in the form of a cloud of dust in air, is likely to occur in normal operation occasionally.

Zone 22 An area in which an explosive dust atmosphere, in the form of a cloud of combustible dust in air is not likely to occur in normal operation but, if it does occur, will persist for a short period only.

Note: for the purpose of this standard, an area or place is a three-dimensional region or space.

P5A.3.2 Flammable Substances

The gas groups and temperature classes for the flammable substances that most often give rise to a potentially explosive atmosphere in a typical water reclamation works with processes that form part of the facilities proposed at TWRP are as follows:

Substance	Gas Group	Temperature Classification
Diesel	IIA	T3
Methane	IIA	T1
Hydrogen Sulphide	IIB	T3
Volatile Organic Compounds*	IIB	T4
Liquified Petroleum Gas	IIA	T1

*The exact composition of the VOC is not able to be determined; the compound detected with the most severe Gas Group and Temperature Classification are used (Ethyl Methyl Ether)

P5A.3.3 Flammable Dusts

No materials that form part of the facilities proposed at TWRP, have been identified having the potential to give rise to potentially explosive atmospheres due to dust.

P5A.3.4 Grades of Release

There are three basic grades of release, as listed below in order of decreasing frequency and likelihood of the explosive gas atmosphere being present:

- Continuous grade: release which is continuous or is expected to occur frequently or for long periods
- Primary grade: release which can be expected to occur periodically or occasionally during normal operation
- Secondary grade: release which is not expected to occur in normal operation and, if it does occur, is likely to do so only infrequently and for short periods

A source of release may give rise to any one of these grades of release, or to a combination of more than one. Examples of the types of releases and the assigned grades are as follows:

Electrical Apparatus for Explosive Gas Atmospheres EN 60079-10: 2003 Annex A and B (informative)	
Sources giving a Continuous Grade of Release:	A: The surface of a flammable liquid in a fixed roof tank, with a permanent vent to atmosphere. B: The surface of a flammable liquid which is open to the atmosphere continuously or for long periods e.g. oil/water separated.
Sources giving a Primary Grade of Release:	A: Seals of pumps, compressors or valves if release of flammable material during normal operation is expected. B: Water drainage points on vessels which may contain flammable liquids, which may release flammable material into the atmosphere while draining off water during normal operation. C: Sample points which are expected to release flammable material into the atmosphere during normal operation. D: Relief valves, vents and other openings which are expected to release flammable material to atmosphere during normal operation.
Sources giving a Secondary Grade of Release:	A: Seals of Pumps, Compressors or valves where release of flammable material during normal operation is not expected. B: Flanges, connections and pipe fittings where release flammable material during normal operation is not expected. C: Sample points which are not expected to release flammable material into the atmosphere during normal operation. D: Relief valves, vents and other openings which are not expected to release flammable material to atmosphere during normal operation.

P5A.3.5 Ventilation

Gas or vapour released into the atmosphere can be diluted by dispersion or diffusion into the air until its concentration is below the lower explosive limit. In TWRP design ventilation is provided by two means:

- (a) Mechanical Ventilation; and
- (b) Natural ventilation.

P5A.3.6 Degree of Ventilation

The following three Degrees of Ventilation have been used in this assessment:

- High:** Can reduce the concentration, at the source of release virtually instantaneously, resulting in a concentration below the LEL.
- Medium:** Can control the concentration, resulting in a stable situation in which the concentration beyond the zone boundary is below the LEL whilst the release is in progress and where the explosive atmosphere does not persist after the release has stopped.
- Low:** Cannot control the concentration whilst release is in progress and / or cannot prevent an undue persistence of a flammable atmosphere after the release has stopped.

P5A.3.7 Availability of Ventilation

The availability of ventilation has an influence on the presence or formation of an explosive gas atmosphere and thus also on the type of zone.

The following three levels of Availability of Ventilation have been used in this assessment:

- Good:** Ventilation is present almost continuously.
- Fair:** Ventilation is expected to be present during normal operation. Discontinuities are permitted provided they occur infrequently and for short periods.
- Poor:** Ventilation which does not meet the standard of good or poor, but discontinuities are not expected to occur for long periods.

Where mechanical ventilation is provided, if it incorporates duty and standby fans it has been assessed as **Good**.

P5A.3.8 Openings Classification

Openings are classified, as per IEC60079-10, as A, B, C and D with the following characteristics:

- Type A** Openings not conforming to the characteristics specified for types B, C or D
- Examples:
- Open passages for access or utilities. Examples of utilities include ducts or pipes through walls, ceilings and floors;
 - Openings which are frequently opened;
 - Fixed ventilation outlets in rooms, buildings and similar openings of types B, C and D which are opened frequently or for long periods.
- Type B** Openings which are normally closed (for example, automatic closing), infrequently opened and close-fitting.
- Type C** Openings which are normally closed (for example, automatic closing), infrequently opened and fitted with sealing devices (for example, a gasket) along the whole perimeter; or two type B openings in series, having independent automatic closing devices.

Type D	Openings normally closed conforming to type C which can only be opened by special means or in an emergency. Type D openings are effectively sealed, such as in utility passages (for example, ducts, pipes) or can be a combination of one opening type C adjacent to a hazardous area and one opening type B in series.
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P5A.4 MARKING OF HAZARDOUS AREAS

Signs are to be displayed at the point of entry to an area where an explosive atmosphere may exist, usually referred to as “Zoned” or “Hazardous” areas.

The sign is triangular in shape with black letters on a yellow background with black edging, as shown in *Figure 4.1*.



Figure 4.1 - Hazardous Area Warning Sign

The purpose of the sign is to warn of entry into zoned or hazardous areas where an explosive atmosphere may be present, so that they can take the necessary precautions in relation to the risk.

The 'Ex' signs are used to:

- Identify places where special workplace or site rules apply e.g. no smoking, antistatic footwear to be worn, or access restricted to authorised people;
- Identify where portable or mobile equipment must be of an explosion protected design e.g. hand torches, vehicles or cleaning machines; and
- Identify for the purposes of audit or later plant modifications, where fixed equipment must be of an explosion-protected design.

The 'Ex' warning sign, should be positioned at the points of entry to a hazardous area. If the hazardous area is located in places that do not have obvious entry points, for example in an open area around multiple zoned installations. The risk area must be identified and its presence communicated by additional means (e.g. clear descriptive text on a sign below the 'Ex' warning sign or marked up site drawing near the 'Ex' warning sign).



Figure 4-2: Example of Descriptive Signage

P5A.5 HAZARDOUS AREA REGISTER

This HAC is based on the reference design presented herein at the time of tender, which describes minimum criteria, which are to be developed by the Contractor to ensure compliance with FSSD and DfS requirements. Review will be required to assess any adjustments needed due to the particular equipment selected by the Contractor.

Process Plant Area	Source of Release		Level of Release (Continuous, Primary, Secondary)	Hazardous Material (Liquid, Gas, Solid)	Ventilation			Hazardous Area Zone 0,1,2,NH	Comments
	Description	Location			Type	Degree	Availability		
Thermal Hydrolysis Plant (WSD11)	Thermal Hydrolysis Plant	External to pipes and vessels	Secondary	Gas – Methane; Hydrogen Sulphide	Mechanical (OC)	Medium	Good	NH	To be reviewed when final layout established
	Thermal Hydrolysis Plant	Safety Pressure Relief Valves (SRV)	Primary	Gas – Methane; Hydrogen Sulphide	Natural	High	Good	Zone 1: a radius of 0.5m around the SRV due to minor leakages Zone 2: a radius of 5m around the SRV due to high pressure relief	Based on weighted type over pressure relief valve

Process Plant Area	Source of Release		Level of Release (Continuous, Primary, Secondary)	Hazardous Material (Liquid, Gas, Solid)	Ventilation			Hazardous Area Zone 0,1,2,NH	Comments
	Description	Location			Type	Degree	Availability		
DWAS Silos (WSB01)	Live Bottom Silo	Above Covers (Type B)	Secondary	Gas – Methane; Hydrogen Sulphide	Mechanical	Medium	Good	NH (Zone 2 during Maintenance- 1m in all directions from openings)	NOTE: This Row is a duplicate of the row in the Biosolids and Digester section of the Register and should be deleted on contract completion Non-hazardous area above the covers due to mechanical ventilation (under covers) and dispersion. However, during maintenance work a Zone 2 with a 1 m minimum extent should be placed around openings, vents and access covers.
	Live Bottom Silo	Under Covers	Primary	Gas – Methane; Hydrogen Sulphide	Mechanical	Medium	Good	Zone 2	NOTE: This Row is a duplicate of the row in the Biosolids and Digester section of the Register and should be deleted on contract completion

Legend:

Location Cover Type: A, B, C, D – refer BS EN 60079-10-1:2009, Annex A; **Hazardous Area**: NH = Non-hazardous, NE = Negligible Extent

NA = Not Applicable; TBC = To Be Confirmed

Process Plant Area	Source of Release		Level of Release (Continuous, Primary, Secondary)	Hazardous Material (Liquid, Gas, Solid)	Ventilation			Hazardous Area Zone 0,1,2,NH	Comments
	Description	Location			Type	Degree	Availability		
Sidestream Nitrogen Removal Facility (WST02)	Solids Separation/ Equalisation Tank	Above Covers (Type B)	Secondary	Gas – methane; Hydrogen Sulphide	Mechanical	High (6 ACPH)	Good	NH (Zone 2 during Maintenance-1m in all directions from openings)	Non-hazardous area above the covers due to mechanical ventilation and dispersion. However, during maintenance work a Zone 2 with a 1 m minimum extent should be placed around openings, vents and access covers.
	Solids Separation/ Equalisation Tank	Under Covers	Primary	Gas – methane; Hydrogen Sulphide	Mechanical	Medium	Good	Zone 2	
	Bioreactor Tanks	Above Covers (Type B)	Secondary	Gas – methane; Hydrogen Sulphide	Natural (open to atmosphere)	High	Good	NH	Non-hazardous area due extent of potential leakage and to mechanical ventilation (under covers) and dispersion.
	Bioreactor Tanks	Under Covers	Primary	Gas – methane; Hydrogen Sulphide	Mechanical	Medium	Good	Zone 2	
	Pumping Station Wet Wells	Under covers	Primary	Gas – methane; Hydrogen Sulphide	Mechanical	Medium	Good	Zone 2	

Process Plant Area	Source of Release		Level of Release (Continuous, Primary, Secondary)	Hazardous Material (Liquid, Gas, Solid)	Ventilation			Hazardous Area Zone 0,1,2,NH	Comments
	Description	Location			Type	Degree	Availability		
	Pumping Station Wet Wells	Above covers (Type B)	Secondary	Gas – methane; Hydrogen Sulphide	Natural (open to atmosphere)	High	Good	NH	
	Valve Chambers	Under covers	Secondary	Gas – methane; Hydrogen Sulphide	Natural	Poor	Low	Zone 2	Any entry into this area will require ventilation and atmosphere monitoring as per site SOP/PTW
	Sludge Tanks	Under covers	Primary	Gas – methane; Hydrogen Sulphide	Mechanical	Medium	Good	Zone 2	
	Sludge Tanks	Above covers (Type B)	Secondary	Gas – methane; Hydrogen Sulphide	Natural (open to atmosphere)	High	Good	NH (Zone 2 during Maintenance-1m in all directions from openings)	Non-hazardous area above the covers due to mechanical ventilation (under covers) and dispersion. However, during maintenance work a Zone 2 with a 1 m minimum extent should be placed around openings, vents and access covers.
	Effluent Tanks	Under covers	Primary	Gas – methane; Hydrogen Sulphide	Mechanical	Medium	Good	Zone 2	

Process Plant Area	Source of Release		Level of Release (Continuous, Primary, Secondary)	Hazardous Material (Liquid, Gas, Solid)	Ventilation			Hazardous Area Zone 0,1,2,NH	Comments
	Description	Location			Type	Degree	Availability		
	Effluent Tanks	Above covers (Type B)	Secondary	Gas – methane; Hydrogen Sulphide	Natural (open to atmosphere)	High	Good	NH (Zone 2 during Maintenance-1m in all directions from openings)	Non-hazardous area above the covers due to mechanical ventilation (under covers) and dispersion. However, during maintenance work a Zone 2 with a 1 m minimum extent should be placed around openings, vents and access covers.

Legend:

Location Cover Type: A, B, C, D – refer BS EN 60079-10-1:2009, Annex A; **Hazardous Area**: NH = Non-hazardous, NE = Negligible Extent

NA = Not Applicable; TBC = To Be Confirmed